The Monitor

Let's get to know the computer monitor better first. The most important human interface device in a computer is the monitor, because it is what we humans look at to understand what is currently happening inside our computer system. Even though the computer has no use for text and graphics and works only in terms of ones and zeroes (data), the computer display shows us text and graphics, which our minds are able to understand.

Most of us work with Cathode Ray Tube (CRT) monitors, which are much like the TVs we all have at home. A few of us choose to pay that little extra and opt for Liquid Crystal Display (LCD) monitors (as on laptops).

The words we most often hear when dealing with monitors are "refresh rate" and "resolution". Here, maximum resolution is the maximum number of dots (pixels) that a monitor can display along its horizontal axis and vertical axis. Thus, a monitor with a maximum resolution of 1024 x 768 can display a maximum of 1024 pixels along its horizontal axis and 768 pixels along its vertical axis.

The refresh rate of a monitor is pretty straightforwardly, the number of times it can draw a whole screen of pixels (1024 horizontal and 768 vertical, in the previous example) per second. So a refresh rate of 85 Hertz (Hz, or number of times per second) at a resolution of 1024 x 768 means that a monitor is drawing the whole screen of 1024 x 768 pixels 85 times per second.

Another term you will come across is wide-screen. In order to explain wide-screen, we first have to talk about aspect ratios:

The aspect ratio of a display is the ratio of the number of horizontal pixels to vertical pixels. The most common aspect ratio is 4:3 for most computer monitors. However, in order to display certain games and most DVD movies optimally, the wide-screen display was made.